

E029

BSCNE Capstone Project

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A capstone task asks you to create a subnet for a lab of 50 hosts. The lab is on the second floor and uses gigabit switches. The rest of the company uses /24 subnets. What is the smallest subnet that fits?

- A. /27
- B. /25
- C. /26
- D. /24

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

How many usable host addresses does a /28 subnet provide?

- A. 14
- B. 28
- C. 4
- D. 16

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

You place an office PC in VLAN 10 and a printer in VLAN 20 on the same 48-port switch. The switch was installed last month and has spare fiber uplinks. The PC cannot reach the printer. What is missing?

- A. Put both in the same subnet but keep separate VLANs.
- B. Inter-VLAN routing through a router or Layer 3 interface.
- C. A fiber uplink connection.
- D. Nothing; devices on one switch always reach each other.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A rule list reads, in order: (1) permit HTTPS from any source to web server 10.0.5.10; (2) deny all traffic from 10.0.9.0/24; (3) implicit deny. The 10.0.9.0/24 range belongs to a guest Wi-Fi added last quarter. A guest at 10.0.9.20 opens HTTPS to 10.0.5.10. What happens?

- A. Denied, because guest networks are always blocked.
- B. Permitted, because rule 1 matches before rule 2 is checked.
- C. Denied by the implicit deny.
- D. Denied, because the more specific deny rule overrides.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Using the same rule list, a host at 10.0.7.15 tries SSH to 10.0.5.10. What happens?

- A. Permitted, since unmatched traffic is allowed.
- B. Permitted by rule 1.
- C. Denied by the implicit deny.
- D. Denied by rule 2.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A task asks you to let a reporting service read files in one cloud storage bucket. A teammate suggests giving it the account administrator role so nothing breaks during grading. The bucket holds 3 GB of CSV files. What should you grant?

- A. Access depends on the file size.
- B. Account administrator, to avoid errors.
- C. Read and write on all buckets.
- D. A role with read-only access to that one bucket.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Your deployment script creates a new virtual network each time it runs and has the production database password typed into it. The script is 140 lines long and was tested on a laptop. The evaluator will run it in a fresh environment. What is the best improvement?

- A. Shorten the script.
- B. Make it declarative/idempotent and move secrets and environment values into parameters or a secrets store.
- C. Run it only once so duplicates never occur.
- D. Add more hard-coded values so it works the same everywhere.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A declarative deployment has already built the environment successfully. What should happen if it is run again with no changes?

- A. It builds a second copy.
- B. It fails, since resources exist.
- C. It deletes the environment.
- D. It reports no changes and leaves resources as they are.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

You applied a policy that should allow web traffic to a server and block remote-desktop access from the internet. The configuration saved with no errors, and the console shows a green checkmark. The server has 16 GB of RAM. How should you confirm the task before submitting?

- A. Trust the green checkmark.
- B. Add more RAM to be safe.
- C. Test only that the web page loads.
- D. Test from an outside client: web should load and remote desktop should be refused.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

On a stateless network filter, you allowed inbound TCP 443 to a web subnet but wrote no outbound rules. The subnet has 12 servers built from the same image. Browsers time out. What rule is missing?

- A. An outbound rule allowing replies to clients on ephemeral ports.
- B. An inbound rule for port 80.
- C. Rebuild the servers from a new image.
- D. Nothing; replies are allowed automatically.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

On a stateful firewall, inbound HTTPS to a server is allowed. What rule is needed for the server's replies?

- A. A rule allowing all outbound traffic.
- B. An inbound rule for ephemeral ports.
- C. None; replies to allowed sessions pass automatically.
- D. An outbound rule for ephemeral ports.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

You configure a router interface for a task, test it, and it works. The router is in a lab rack with blue cables. Overnight the lab loses power, and in the morning your changes are gone. What step was missed?

- A. Saving the running configuration so it loads at startup.
- B. Using different cables.
- C. Nothing; routers always keep changes after power loss.
- D. Testing the change.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A capstone task asks you to create a subnet for a lab of 50 hosts. ~~The lab is on the second floor and uses gigabit switches. The rest of the company uses /24 subnets.~~ What is the smallest subnet that fits?

Target word: smallest prefix for 50 hosts **Decides:** /26 gives 62 usable; /27 gives only 30.

A	Inverted operation	A larger prefix is a smaller subnet: 30 usable.
B	Adjacent category	Fits (126) but is not the smallest.
C	Correct	$2^6 - 2 = 62$ usable, enough for 50.
D	Camouflage	Matches the company standard, which does not decide the smallest fit.

2. Answer A Concept 1 • transfer

How many usable host addresses does a /28 subnet provide?

Target word: solve for hosts from prefix **Decides:** $2^{(32 - 28)} - 2$.

A	Correct	$16 - 2 = 14$.
B	Wrong stem number	The prefix length, not a host count.
C	Inverted operation	That is the number of host bits, not hosts.
D	Dropped element	Forgets the network and broadcast addresses.

3. Answer B Concept 2

You place an office PC in VLAN 10 and a printer in VLAN 20 on the same 48-port switch. ~~The switch was installed last month and has spare fiber uplinks.~~ The PC cannot reach the printer. What is missing?

Target word: why VLANs cannot talk **Decides:** Different VLANs need Layer 3 routing.

A	Confusion set	Mismatches VLAN and subnet design; still no routing.
B	Correct	VLANs are separate broadcast domains.
C	Camouflage	Uplinks do not route between VLANs.
D	Absolute language	VLANs separate them by design.

4. Answer B Concept 3

A rule list reads, in order: (1) permit HTTPS from any source to web server 10.0.5.10; (2) deny all traffic from 10.0.9.0/24; (3) implicit deny. The 10.0.9.0/24 range belongs to a guest Wi-Fi ~~added last quarter~~. A guest at 10.0.9.20 opens HTTPS to 10.0.5.10. What happens?

Target word: result of first-match evaluation **Decides:** Rule 1 matches first.

A	Absolute language	The rules decide, not the label.
B	Correct	First match wins.
C	Adjacent category	The implicit deny applies only when nothing matches.
D	Inverted operation	Order, not specificity, decides.

5. Answer C Concept 3 • transfer

Using the same rule list, a host at 10.0.7.15 tries SSH to 10.0.5.10. What happens?

Target word: traffic that matches no explicit rule **Decides:** Not HTTPS, not from 10.0.9.0/24.

A	Inverted operation	Unmatched traffic is denied.
B	Confusion set	Rule 1 permits only HTTPS.
C	Correct	No explicit rule matches.
D	Wrong stem number	The source is 10.0.7.15, not 10.0.9.x.

6. Answer D Concept 4

A task asks you to let a reporting service read files in one cloud storage bucket. A teammate suggests giving it the account administrator role so nothing breaks during grading. ~~The bucket holds 3 GB of CSV files.~~ What should you grant?

Target word: correct permission scope **Decides:** Read-only on one bucket.

A	Camouflage	Size does not decide permissions.
B	Inverted operation	Far broader than needed.
C	Adjacent category	Adds write and scope not requested.
D	Correct	Least privilege.

7. Answer B Concept 5

Your deployment script creates a new virtual network each time it runs and has the production database password typed into it. ~~The script is 140 lines long and was tested on a laptop.~~ The evaluator will run it in a fresh environment. What is the best improvement?

Target word: automation quality **Decides:** Not idempotent; secret hard-coded.

A	Camouflage	Length is not the flaw.
B	Correct	Safe reruns and no embedded secrets.
C	Dropped element	Avoids the symptom, keeps the flaw.
D	Inverted operation	Makes portability worse.

8. Answer D Concept 5 • transfer

A declarative deployment has already built the environment successfully. What should happen if it is run again with no changes?

Target word: idempotency **Decides:** State already matches.

A	Inverted operation	That is non-idempotent behavior.
B	Confusion set	Imperative create scripts fail this way; declarative ones do not.
C	Adjacent category	Deletion requires removing it from the code.
D	Correct	Idempotent by design.

9. Answer D Concept 6

You applied a policy that should allow web traffic to a server and block remote-desktop access from the internet. The configuration saved with no errors, and the console shows a green checkmark. ~~The server has 16 GB of RAM.~~ How should you confirm the task before submitting?

Target word: verification method **Decides:** Evaluators test functionality.

A	Dropped element	Saving is not the same as working.
B	Camouflage	Memory does not verify rules.
C	Adjacent category	Skips the blocked path.
D	Correct	Proves both the allowed and blocked paths.

10. Answer A Concept 7

On a stateless network filter, you allowed inbound TCP 443 to a web subnet but wrote no outbound rules. ~~The subnet has 12 servers built from the same image.~~ Browsers time out. What rule is missing?

Target word: missing rule on a stateless filter **Decides:** Return traffic needs an explicit rule.

A	Correct	Stateless filters do not track sessions.
B	Adjacent category	A different service; replies are the issue.
C	Camouflage	The image does not decide filtering.
D	Confusion set	True only for stateful filters.

11. Answer C Concept 7 · transfer

On a stateful firewall, inbound HTTPS to a server is allowed. What rule is needed for the server's replies?

Target word: reverse: stateful behavior **Decides:** Stateful tracking permits replies.

A	Absolute language	Far broader than required.
B	Confusion set	Wrong direction and not needed.
C	Correct	The firewall tracks the connection.
D	Inverted operation	Needed on stateless filters, not stateful.

12. Answer A Concept 8

You configure a router interface for a task, test it, and it works. ~~The router is in a lab rack with blue cables.~~ Overnight the lab loses power, and in the morning your changes are gone. What step was missed?

Target word: change practice step **Decides:** Running changes were not saved to startup configuration.

A	Correct	Unsaved changes are lost on reboot.
B	Camouflage	Cables do not decide persistence.
C	Absolute language	Only saved configuration survives.
D	Dropped element	It was tested; saving was missed.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Subnet sizing	Usable IPv4 hosts in a traditional subnet = $2^{(32 - \text{prefix})} - 2$ (network and broadcast); choose the smallest prefix that fits the need.	Forgetting the two reserved addresses, or thinking a bigger prefix number means a bigger subnet.	A host count and a list of prefixes.
2	VLAN segmentation	VLANs split one switch into separate broadcast domains; traffic between VLANs needs a Layer 3 device (router or Layer 3 switch).	Expecting hosts in different VLANs to talk just because they share a switch.	Two VLANs on one switch that cannot reach each other.
3	Rule order and implicit deny	Firewall and ACL rules are checked top-down; the first match wins, and traffic that matches nothing hits an implicit deny.	Assuming a later, more specific rule overrides an earlier broad one.	An ordered rule list and a packet to evaluate.
4	Least privilege for cloud resources	Grant each user or service only the permissions and scope its task needs, preferring roles over shared or long-lived credentials.	Granting administrator rights so the task “just works.”	A task that needs narrow access and a broad grant.
5	Idempotent, version-controlled automation	Deployment automation should declare desired state, be safe to rerun, and live in version control with parameters instead of hard-coded values.	Scripts that create duplicates when rerun or embed environment-specific secrets.	A deployment that must run in more than one environment or more than once.
6	Verify like the evaluator	A working deployment is proven by testing end-to-end from the client’s view — both paths that should work and paths that should be blocked.	Checking only that the configuration saved without testing traffic.	A task graded by logging in or testing functionality.
7	Stateful vs. stateless filtering	Stateful filters allow return traffic for permitted sessions automatically; stateless filters need explicit rules for both directions, including ephemeral return ports.	Writing inbound-only rules on a stateless filter.	Requests arrive but replies never return.
8	Safe change practice	Back up the current configuration, plan a rollback, make the change, verify, and save the configuration so it survives a restart.	Changing a live device with no backup or forgetting to save.	A change that disappears after a reboot, or a change that breaks access.

Course scope (public description): In the BSCNE Capstone Project, students choose from a fixed set of tasks that they might encounter in the workplace. Examples might include configuring network components, applying security policies to a set of cloud resources, or automating a network deployment using orchestration tools. Students will submit their tasks for evaluation in a working, deployed environment where an evaluator can check their results by inspecting configurations or logging into machines or services to test functionality.